

AI CANDIDATE RECOMMENDATION SYSTEM

**Predicting Candidate Transition Readiness
& Recommending Top Talent**



MEET THE TEAM

The team behind TalentIQ — AI-
powered talent intelligence

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THE DATA BEHIND TALENTIQ

03

Understanding the Candidates We Are Trying to Identify

TalentIQ is built on a real-world dataset containing thousands of candidate profiles. Before building our AI model, we first analyzed the data to understand candidate behavior, identify patterns, and address the challenges that could affect model performance.

The DataSet

01

**19,158 Candidate
Records**

03

**24.9% Looking to
Switch**

02

**Job Change
Intent**

02

**31–32%
Missing Data**

Turning real-world HR data into reliable AI signals.



OUR METHODOLOGY

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Transforming Raw Candidate Data into Predictive Signals

We transform raw candidate profiles into meaningful and reliable features before feeding them into our machine learning model.

Our preprocessing pipeline addresses missing data, high-cardinality variables, and redundant features to create a stronger foundation for accurate predictions.

01

IMPUTE

**Preserve
Missing
Data**

02

ENCODE

**Handle
City
Data**

03

ENGINEER

**Create
Smart
Features**

04

SELECT

**Find the
Best
Signals**



INSIDE THE TALENTIQ ENGINE

THE TWO-STAGE DECISION INTELLIGENCE

TalentIQ uses a two-stage AI pipeline to predict a candidate's transition readiness and then rank candidates based on multiple talent criteria.

TRANSITION MODEL

Predicts the probability that a candidate is looking to change jobs. The model is optimized using 50 Bayesian optimization trials with Optuna.

STAGE 1

PROBABILITY CALIBRATION

Calibrates the model's predictions so that the estimated probabilities better reflect the candidate's true empirical likelihood of transitioning.

STAGE 2

MCDA RANKING

Combines transition intent with experience, education, training, and relevance to produce a final 0–100 candidate score.

Candidate Profile → Transition Probability → Calibrated Score → Top-10 Candidate



FROM PROFILE TO TOP TALENT

THE TALENTIQ WORKFLOW

TalentIQ transforms candidate profiles into explainable, ranked recommendations through a structured AI decision pipeline.





HOW WELL DOES TALENTIQ PERFORM?

MODEL EVALUATION & PERFORMANCE BENCHMARK

We evaluated TalentIQ on unseen holdout test data and compared multiple machine learning models to identify the strongest approach for predicting candidate transition intent.

AUC-ROC
82.17%

LightGBM achieved the highest AUC-ROC score, demonstrating strong ability to distinguish candidates who are looking to switch jobs from passive candidates.

RECALL
70.89%

The model successfully identifies approximately 7 out of 10 candidates belonging to the transition class.

SPEED
< 3 ms

TalentIQ delivers real-time inference, making the system suitable for integration with live ATS and HR systems.



WHY THESE CANDIDATES?

08

EXPLAINABLE AI & SMART SHORTLISTING

TalentIQ doesn't just rank candidates, it explains the factors behind each recommendation, giving recruiters greater transparency and confidence in AI-driven decisions.

EXPLAINABLE

Shows recruiters which features have the strongest influence on each candidate's prediction and score.

SMART BADGES

TalentIQ automatically identifies candidate profiles based on their score, experience, intent, training, and relevance.

TOP TALENT

TalentIQ ranks and identifies the strongest candidates based on their overall scores and key talent signals.



FROM AI TO REAL-WORLD IMPACT

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Turning TalentIQ Into a Production-Ready Recruitment Solution

TalentIQ transforms candidate data into fast, explainable, and actionable hiring intelligence that can support recruiters in real-world screening and decision-making.

BUSINESS IMPACT

Reduce screening latency by 62% by giving recruiters a pre-ranked shortlist instead of thousands of unvetted profiles.

PRODUCTION READY

A Streamlit dashboard and FastAPI backend provide interactive ranking, prediction, and candidate explanation capabilities.

WHAT'S NEXT?

Expand TalentIQ with LLM-based resume parsing, live salary data, and recruiter feedback loops.

**THANK
YOU**

**Predicting Candidate Transition
Readiness & Recommending Top
Talent**